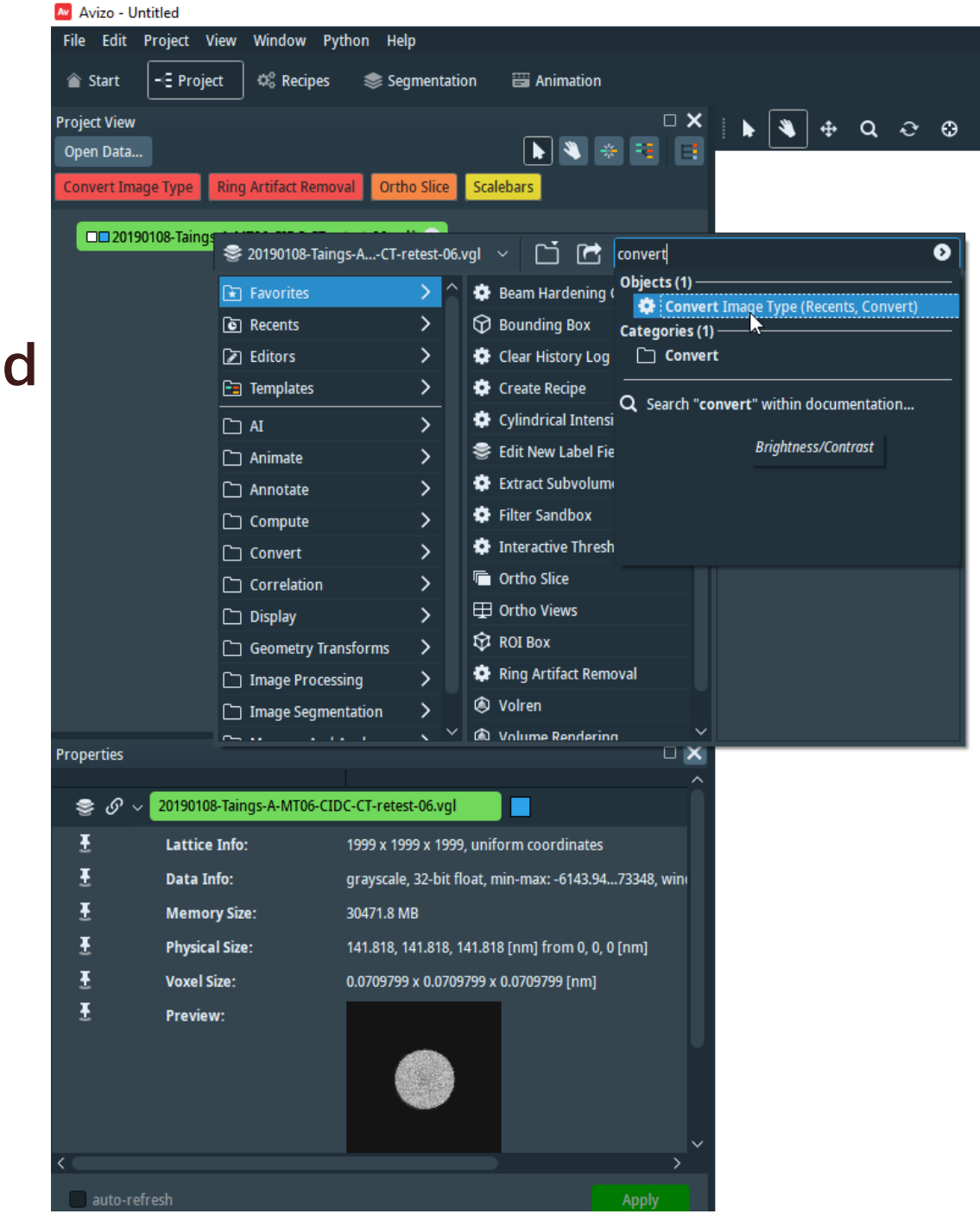
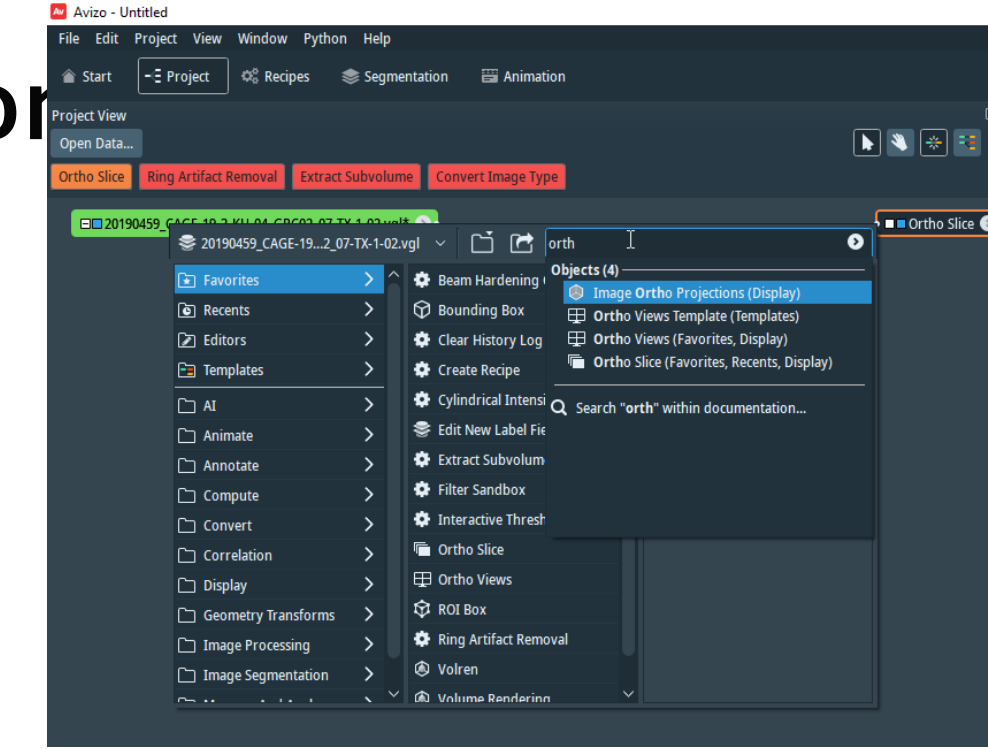
Avizo

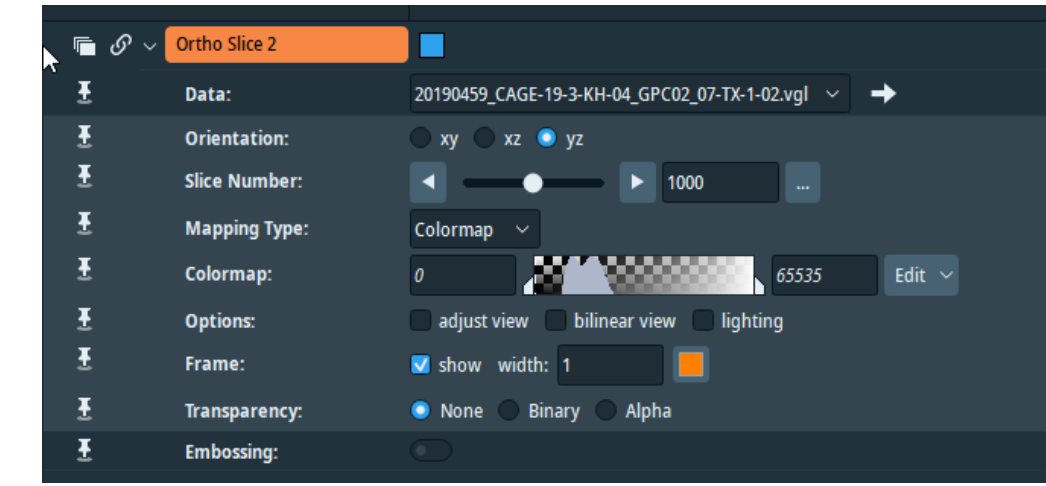
- Drag and drop .vgl file
- Convert to 16 bit unsigned



Processing – visualisation

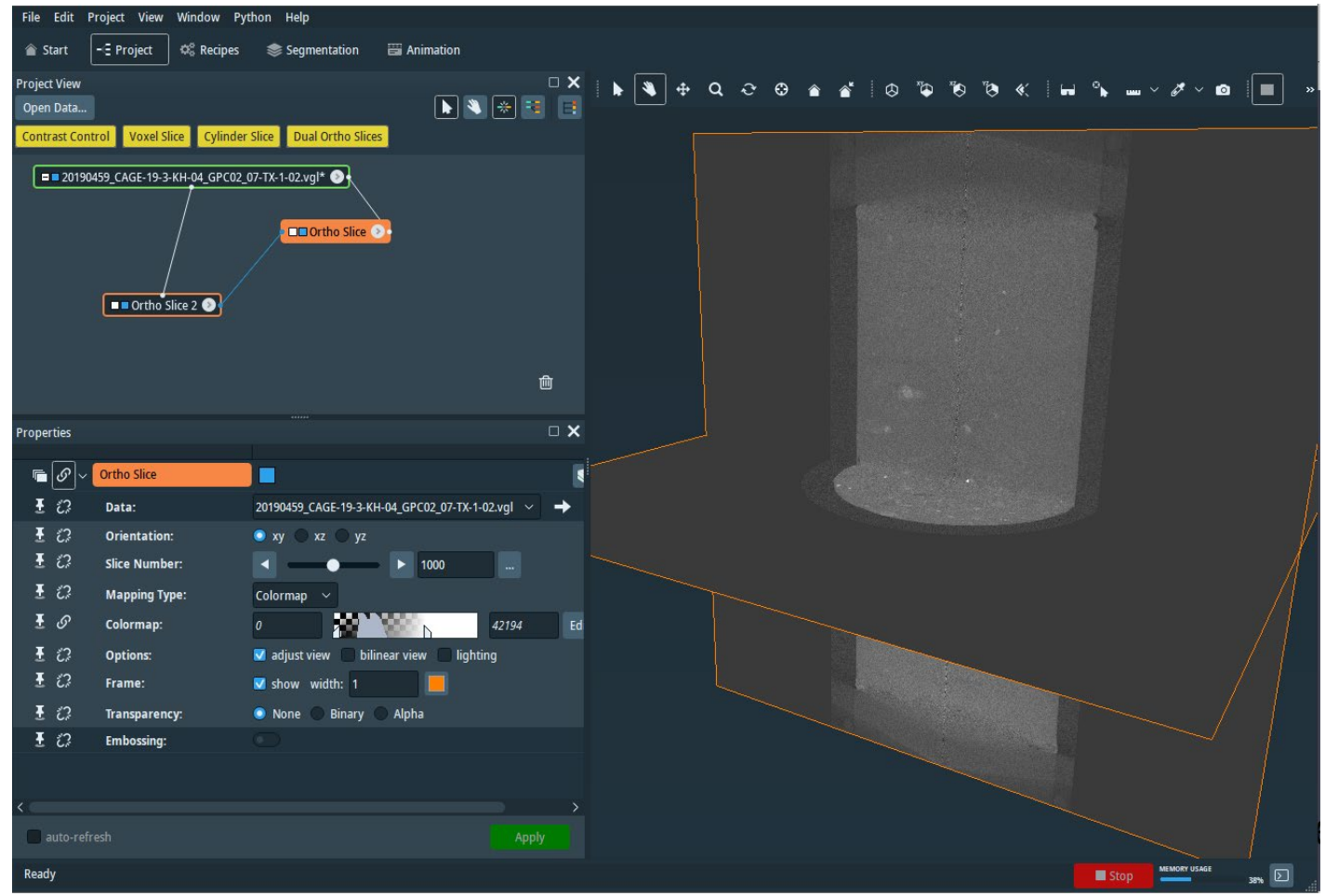
- Once data is loaded, right click on rectangle and select «Ortho slice», either by finding it in the menu, or starting typing «Orth...» and selecting.
- Adjust orientation and colormap in the toolbar bottom left



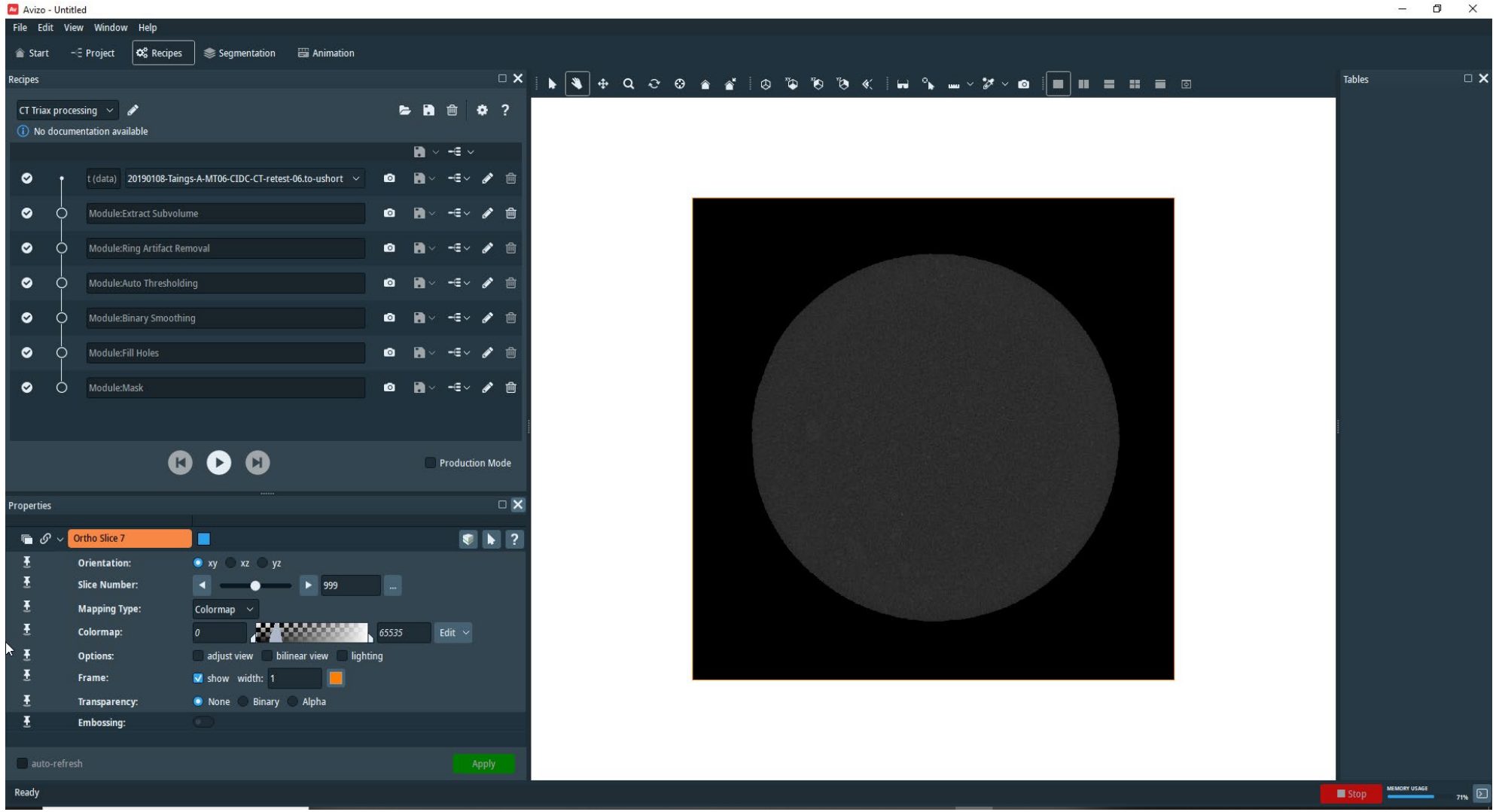


Processing - visualisation

- Here the colormap between the xy and yz slices are linked (clicking and dragging the link next to Colormap to the other ortho slice rectangle.
- Then, when you change the colormap of one, the other changes

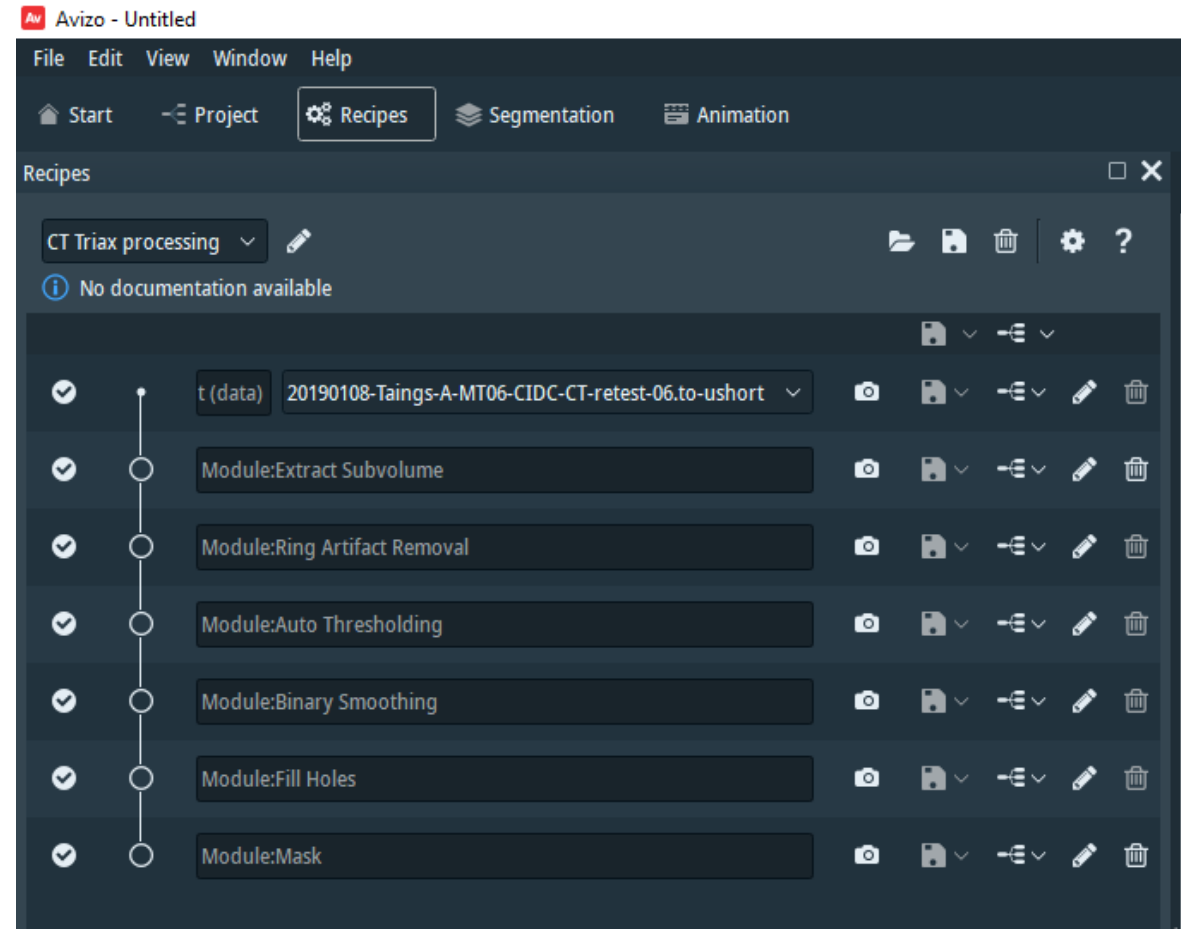


Avizo - Recipe for processing

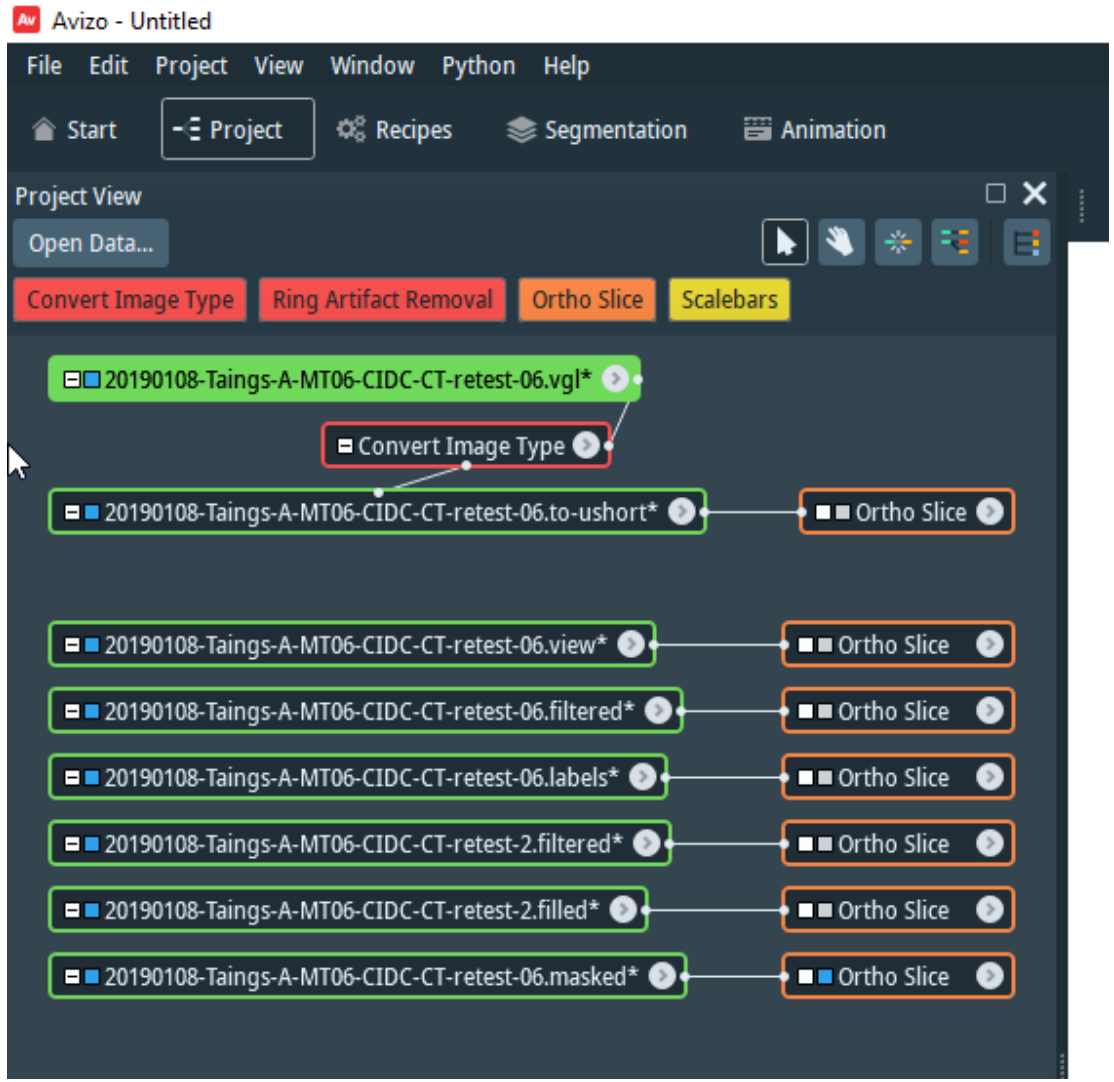


Avizo -

- Recipe extract subvolume
- Ring artefact correction
- Segment the core sample:
  - Autothresholding to make mask
  - Fill holes in the mask
  - Apply the mask to the image

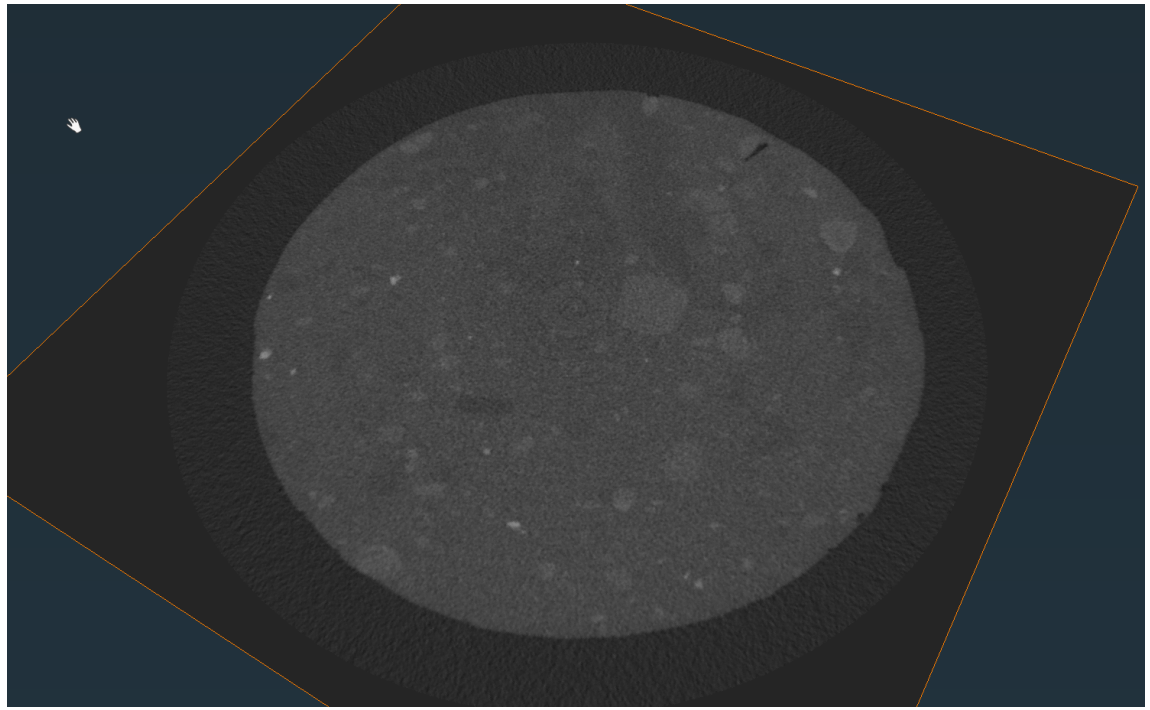


Avizo - Export



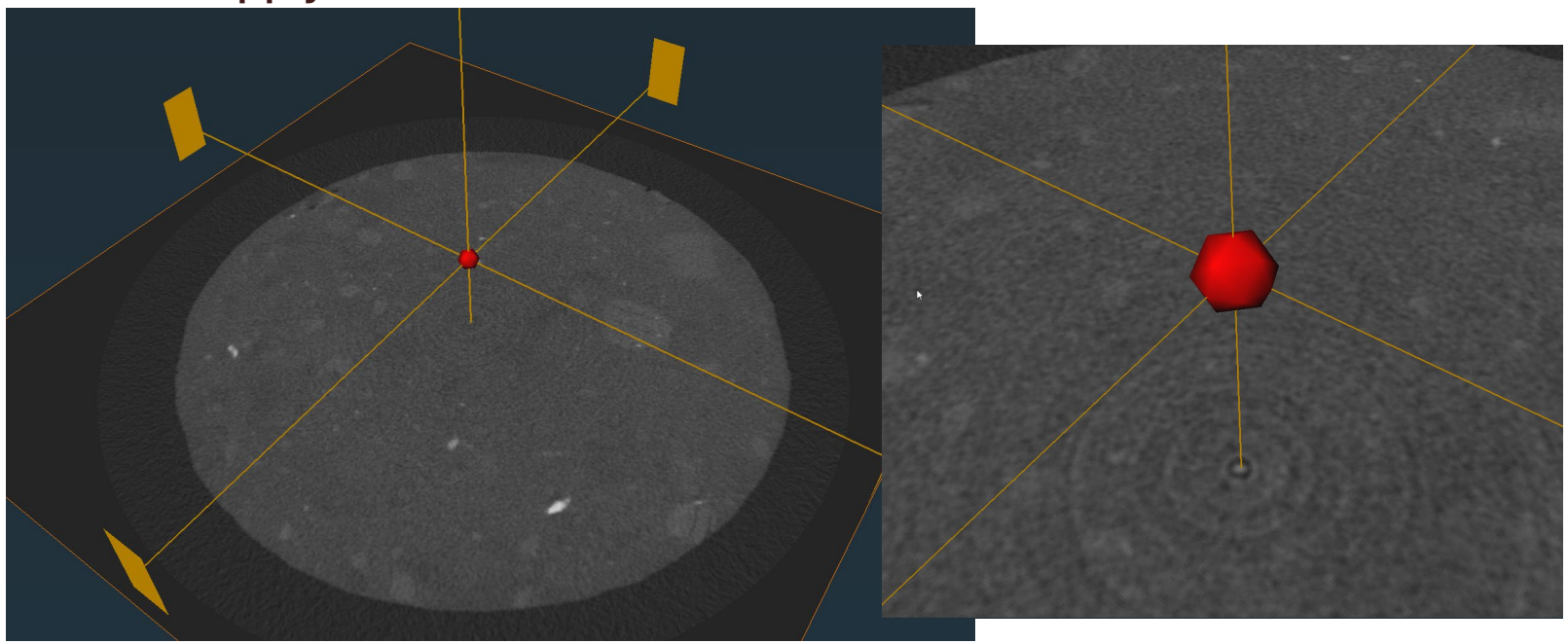
Processing – extract subvolume

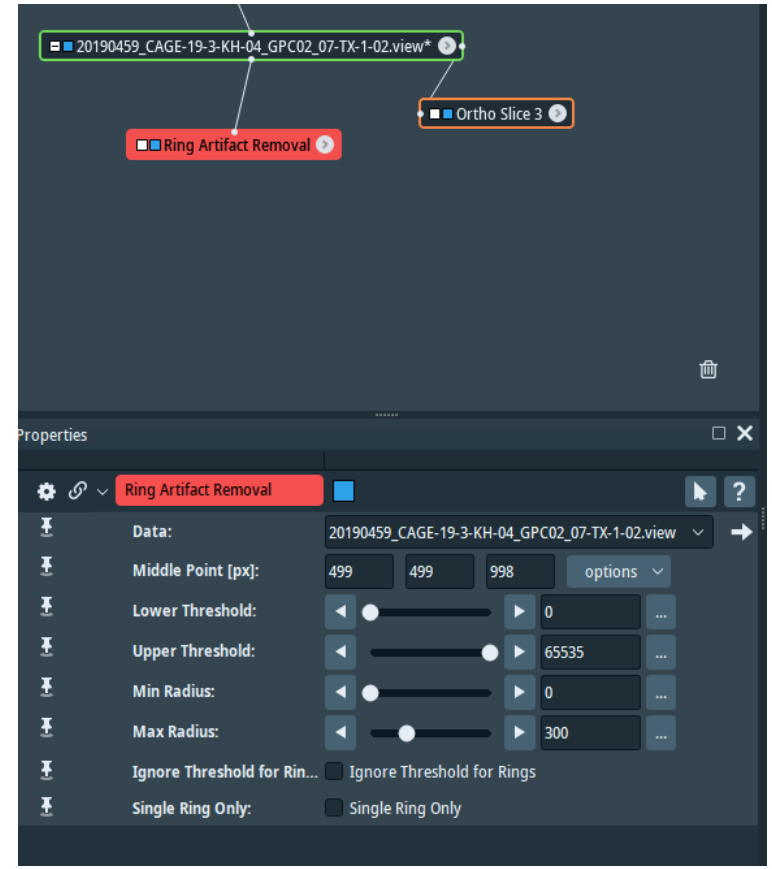
- Using the «extract subvolume» tool and settings shown to the right, click apply.



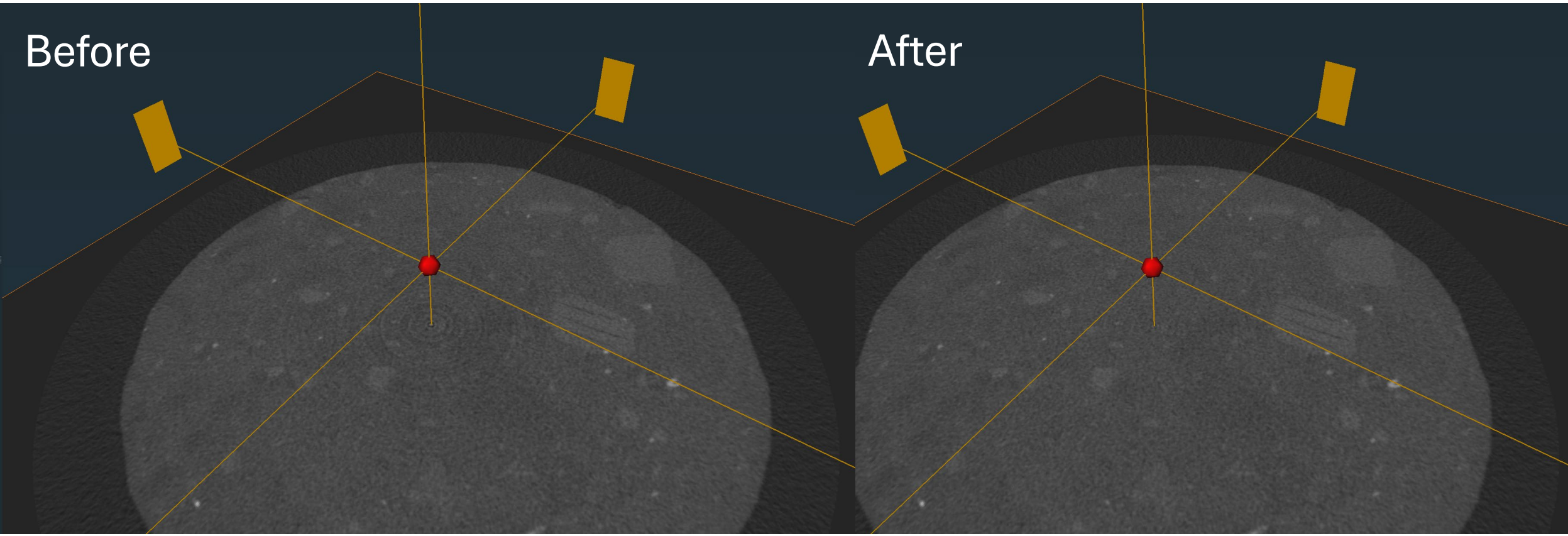
Processing – ring artefact removal

- Next, use the «Ring artifact removal» tool.
- I find 300 max radius works well – if the radius is too high and outside the sample radius, strange things happen.
- With «options», «show dragger» ticked, check that the dragger aligns with the centre of the sample (if the subvolume was selected with the options previous, this should be fine.
- Click apply!

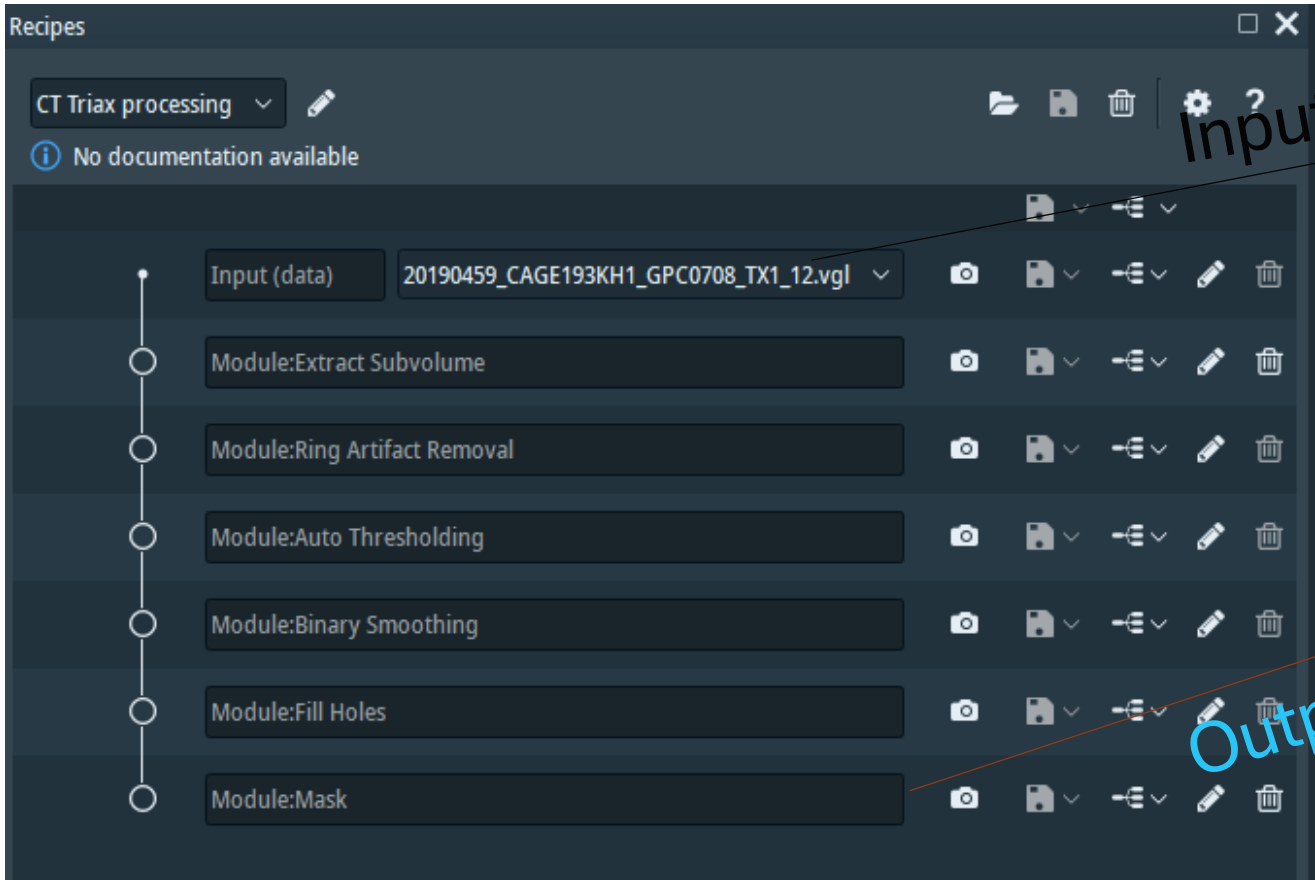


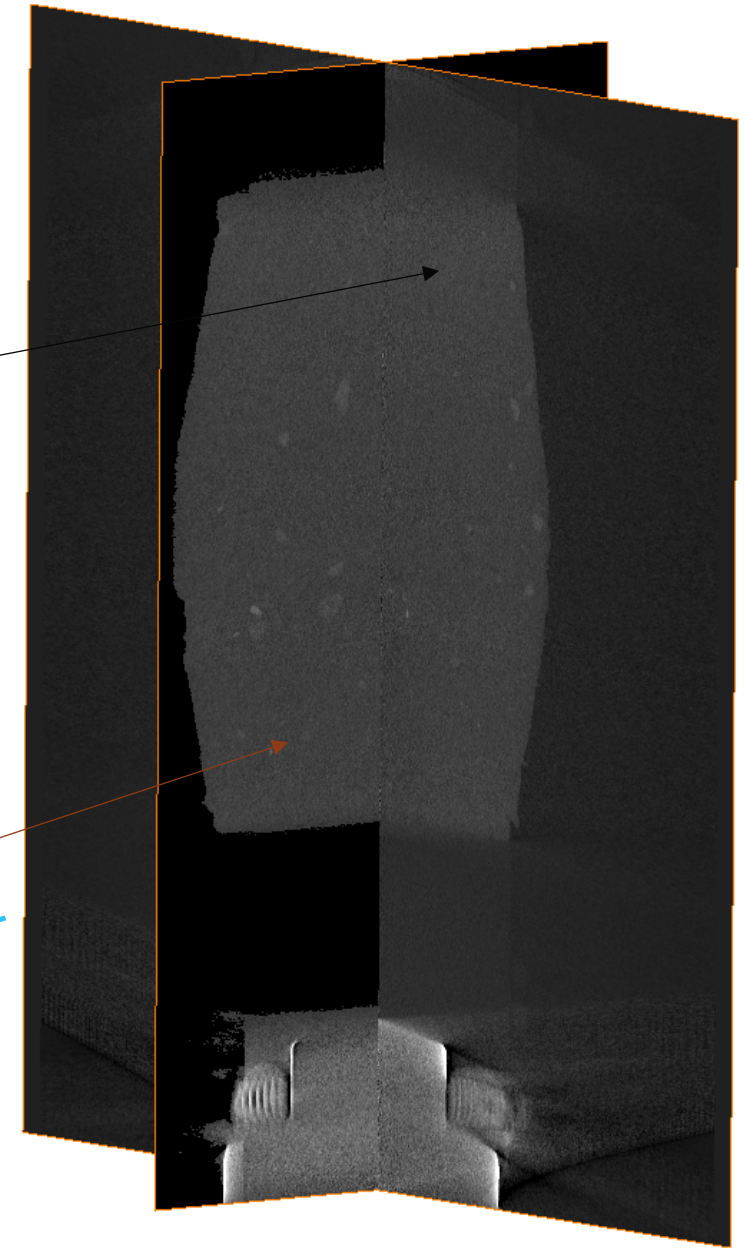


Processing – ring artefact removal



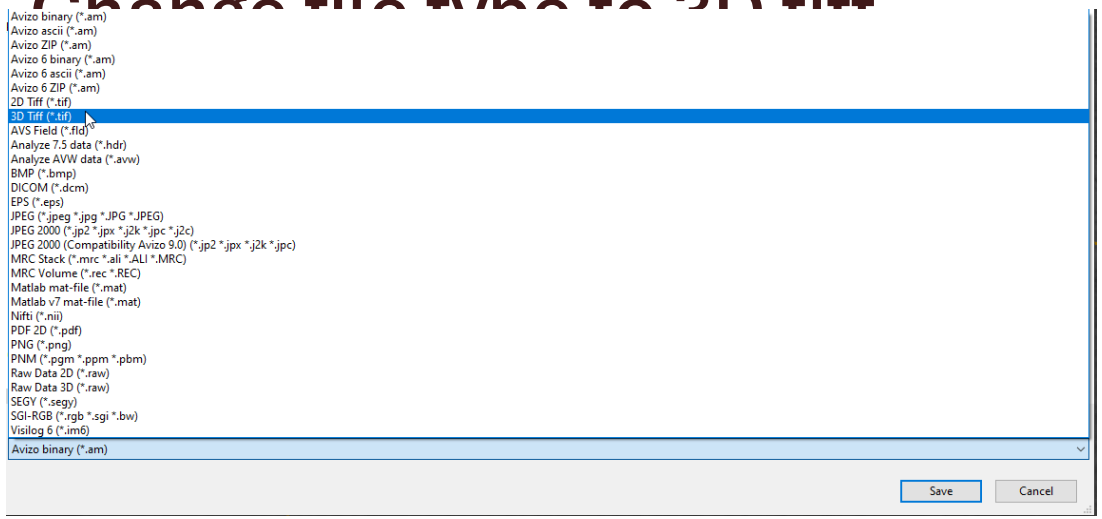
Pre-processing in Avizo

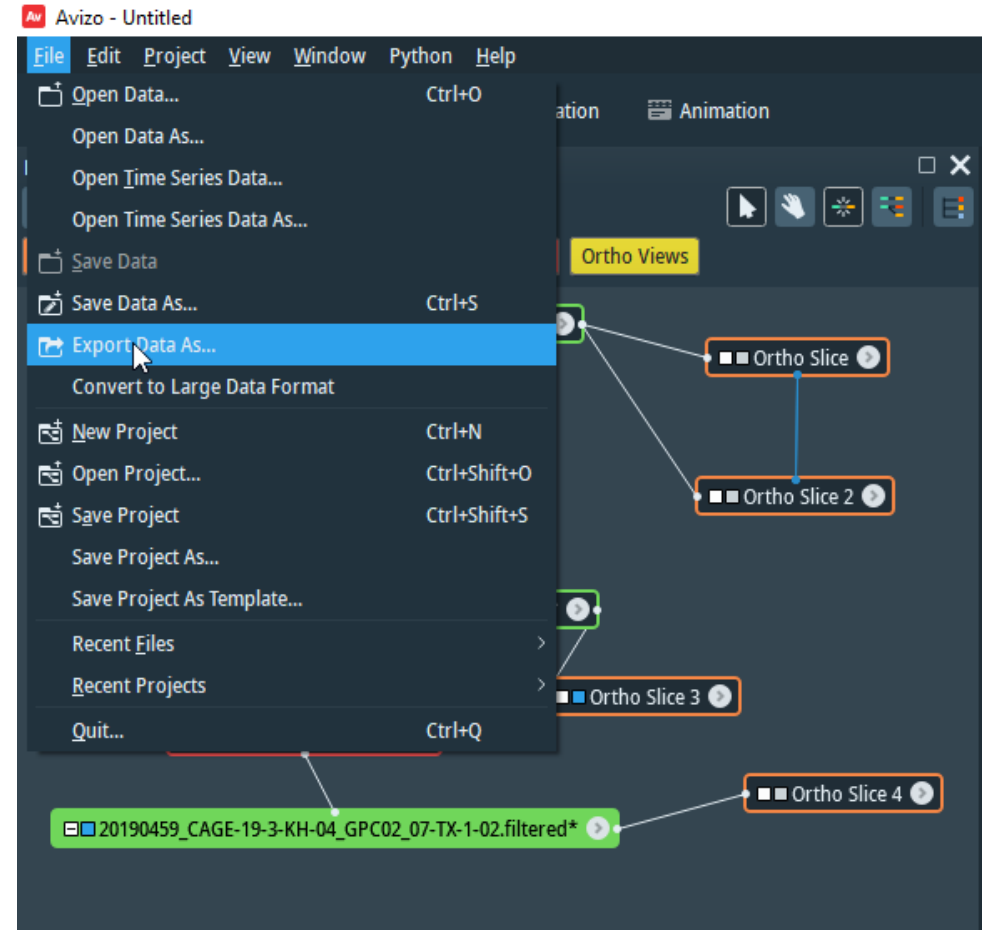
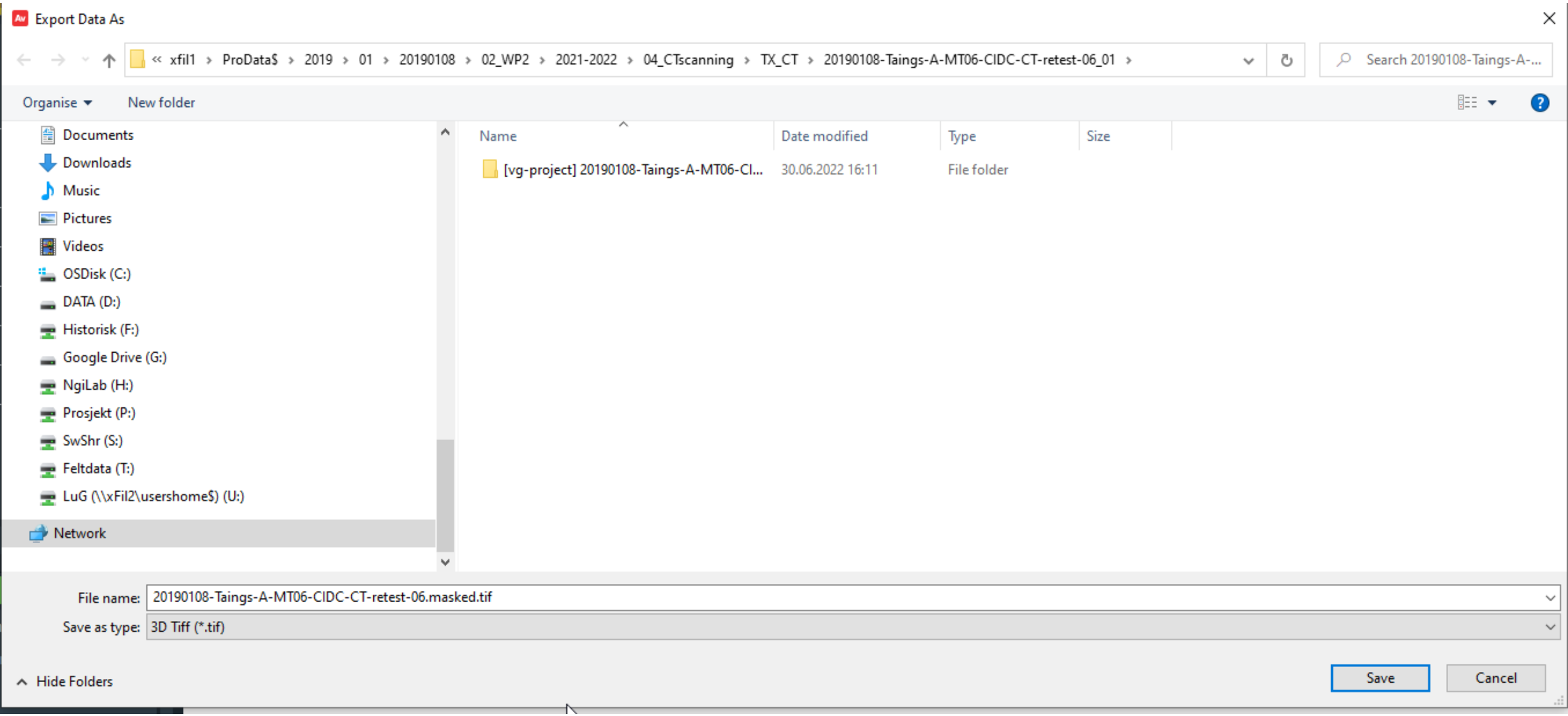




Processing – exporting data

- That's enough in Avizo for now
- Select the data by clicking the corresponding box
- Go to «File», «Export Data As...»
- Change file type to 3D tiff





Export pre-processed image